ModelIt!

Student Activity Pack

Can Nuclear Power Actually Save the Planet?

Modeling Nuclear Fission, Energy Output, and Environmental Trade-offs

Grade 10L1 | NGSS HS-PS1-8, HS-ESS3-2 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you already know about how nuclear power plants produce electricity?

2 Why do you think some people support nuclear power while others oppose it?

3 Draw a diagram showing what you think happens inside a nuclear reactor to produce energy.

4 How does the amount of waste from a nuclear plant compare to the waste from a coal plant producing the same energy?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Uranium Fuel Supply | Fission Rate | Energy Output | Radioactive Waste | Carbon Emissions Avoided

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Full Power

Uranium fuel: Standard load | Fission rate: 100%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Reduced Power

Uranium fuel: Standard | Fission rate: 50%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Fuel Depletion

Full operation until fuel exhaustion

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Cooling Water Demand — The enormous volume of water needed to cool the reactor and condense steam — a typical plant uses 1-2 billion gallons per day, raising concerns about thermal pollution and water scarcity

Fuel Enrichment Level — The percentage of uranium-235 in the fuel — natural uranium is 0.7% U-235, reactor fuel is enriched to 3-5%, and weapons-grade is 90%+, raising proliferation concerns

Public Risk Perception — The level of public fear about nuclear accidents, which is often disproportionate to actual statistical risk but profoundly influences energy policy decisions

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If nuclear power produces almost zero carbon emissions, why is the world still arguing about whether to use it? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the extraordinary energy density of nuclear fuel compared to fossil fuels?**

2. **What does the model reveal about the trade-off between carbon avoidance and waste generation?**

3. **If nuclear power is the lowest-carbon reliable energy source, why do many countries still rely heavily on fossil fuels?**

4. **How should society weigh risks that last decades (climate change) against risks that last millennia (nuclear waste)?**

5. **What assumptions does your model make about waste storage technology and safety?**

8 STEM Challenge: Design a National Energy Portfolio

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YOUR ENGINEERING MISSION

Design a national energy mix for 2050 that meets electricity demand while minimizing both carbon emissions and nuclear waste, balancing cost, reliability, and environmental impact.

THE SCENARIO

The Department of Energy has tasked your team with proposing a 2050 energy portfolio for a nation of 350 million people. You must meet a demand of 4,000 terawatt-hours per year while reducing carbon emissions by 80% from current levels. Nuclear, solar, wind, hydro, and natural gas are all available — but each has trade-offs your model has revealed.

Engineering Design Thinking

1. WHAT IS THE OPTIMAL BALANCE BETWEEN NUCLEAR BASELOAD POWER AND RENEWABLE INTERMITTENT SOURCES?
2. HOW WILL YOU HANDLE THE WASTE GENERATED OVER THE 30-YEAR OPERATING LIFE OF EACH NUCLEAR PLANT?
3. WHAT BACKUP GENERATION WILL YOU INCLUDE FOR WHEN WIND AND SOLAR OUTPUT DROPS?
4. HOW DOES YOUR PORTFOLIO COMPARE TO CURRENT ENERGY MIXES IN TERMS OF COST, RELIABILITY, AND EMISSIONS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Nuclear Engineers design, operate, and maintain nuclear reactors and manage radioactive materials. They work at power plants, national laboratories, the Navy's nuclear fleet, and regulatory agencies, earning $80,000–$150,000/year. Energy Policy Analysts who evaluate energy portfolios for governments earn $65,000–$130,000/year.